



CSCS

Centro Svizzero di Calcolo Scientifico
Swiss National Supercomputing Centre

ETH zürich



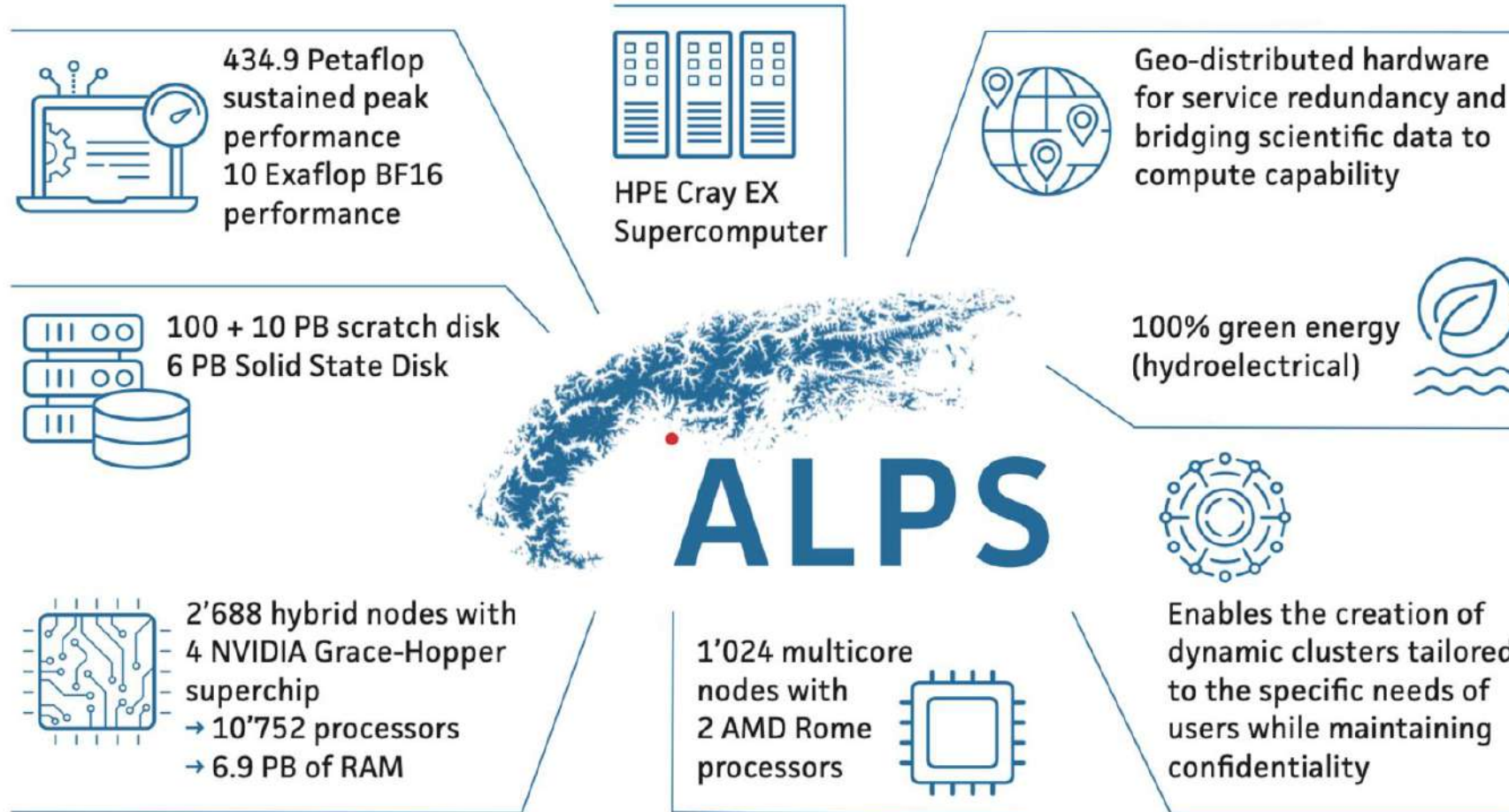
vCluster validation on Alps with ReFrame

HPC Tests Workshop – CUG 2026

April 25th, 2026

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Alps Research Infrastructure



Current locations:

- CSCS
 - Lugano
- EPFL
 - Lausanne
- Paul Scherrer Institute (PSI)
 - Villingen
- ECMWF
 - Bologna
- ±30 vClusters

ReFrame

- Python framework for regression and performance testing on Supercomputers
- Originally developed by a small team in CSCS back in 2016
 - Initially targeted Cray systems
- First public release in May 2017 on Github
 - <https://github.com/reframe-hpc/reframe>
 - 66 contributors, 121 forks, 210 stars
 - 163 releases
- Currently applying to join HPSF
 - <https://hpsf.io/>



ReFrame

- Used by both academic institutions and industry around the world for testing and benchmarking HPC systems
- A few sites known to use ReFrame:
 - CSCS, NVIDIA, EESSI Project, Pawsey, CINECA, SURF, Intel, Excalibur Project, NERSC, ANL, INL, Indiana University, HPE, CDAC, Cornelis Networks, Microsoft Azure, ORNL, DeiC, Univ. of Luxembourg
 - Unofficial and non-exhaustive list
 - You are welcome to answer the ReFrame User Survey if you wish to be on this list :)

Key ReFrame features

■ **Productivity**

- Implements a Python DSL allowing to write tests in a high-level declarative way
- Tests are composable by design and can be extended
- Multi-dimensional test parameterisation

■ **Portability**

- Tests can be reused across HPC systems platforms and different sites
- Auto-detection of processor topology and microarchitecture
- Support for multiple job schedulers, launchers, build systems, module systems, container engines

■ **Performance**

- Efficient resource sharing and dependencies through test fixtures
- Parallel execution of tests
- Performance logging through multiple channels

ReFrame use cases at CSCS

- Maintenance test suites
 - Executed before and after interventions
 - Scheduled maintenances or after *unforeseen events*
 - Daint example: 416 tests
- Production test suites
 - Executed daily
 - Set of single and multi-node (short) jobs
 - Used to track performance trends and regressions
- Work in progress
 - Node validator vService
 - Integration tests executed at node boot time
 - Node vetting
 - GPU burn (dgemm), now looking into using PyTorch (torch-hammer tests from HPE)

Important links

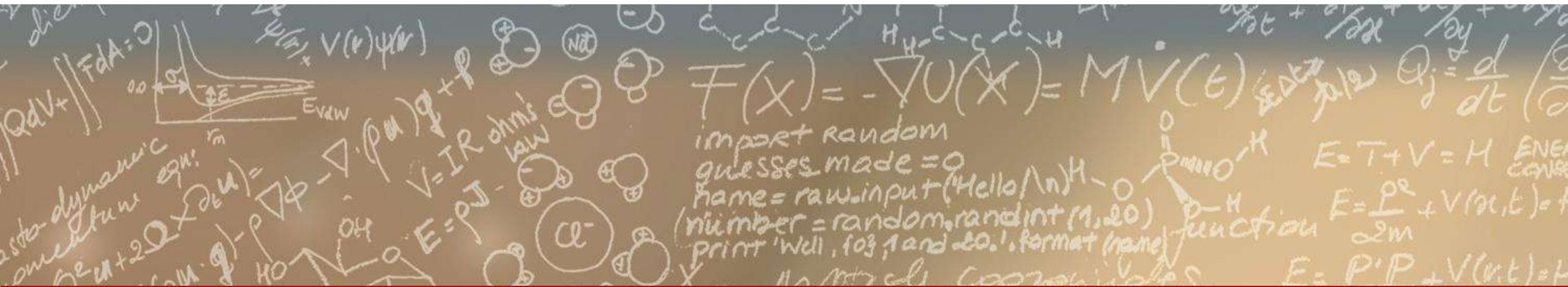
- Git repositories of interest
 - CSCS tests: <https://github.com/eth-cscs/cscs-reframe-tests>
 - Framework: <https://github.com/reframe-hpc/reframe>
- Official documentation: <https://reframe-hpc.readthedocs.io/>
 - Includes link to join the community Slack channel
- ReFrame User Survey 2026
 - <https://tinyurl.com/ReFrameSurvey2026>



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Thank you for your attention.

ReFrame User Survey 2026

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